

matrixdiagrams.sty

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Overview

`matrixdiagrams` draws schematic matrix diagrams (bracketed boxes with diagonals, shaded blocks, labels, and partition lines) directly in TikZ, for typesetting matrix decompositions. Every diagram is built with one command, `\matrixdiagram[scale][aspect]{content}`, where `content` is a sequence of small drawing commands (`\mdFill`, `\mdLabel`, `\mdDiagBand`, ...) working in a normalized coordinate system that automatically adapts to the diagram's scale and aspect ratio. The sections below give a quick reference followed by every command with a runnable example.

1 Quick Reference

<code>\matrixdiagram [s][a]{c}</code>	Main command. <code>s</code> = scale (default 1), <code>a</code> = aspect (default 1). Position args below are fractions $x, y \in [0, 1]$, any aspect.
<code>\mdDiag</code>	Main diagonal
<code>\mdDiagBand</code>	Tridiagonal band.
<code>\mdDiagBandN {n}</code>	n -diagonal band (n odd).
<code>\mdBidiagUpper</code>	Main diagonal + super-diagonal.
<code>\mdBidiagLower</code>	Main diagonal + sub-diagonal.
<code>\mdHessenberg</code>	Upper-triangular outline + sub-diagonal.
<code>\mdHessenberg [color]</code>	Same with filled interior.
<code>\mdDiagFrom {x1}{y1}{x2}{y2}</code>	Main diagonal from (x_1, y_1) to (x_2, y_2)
<code>\mdDiagBandFrom {x1}{y1}{x2}{y2}</code>	Tridiagonal band, clipped.
<code>\mdDiagBandNFrom {n}{x1}{y1}{x2}{y2}</code>	n -diagonal band, clipped.
<code>\mdFill [col]{x0}{y0}{x1}{y1}</code>	Filled rectangle. Default color <code>gray!20</code> .
<code>\mdVLine {x}</code>	Vertical line, full height.
<code>\mdHLine {y}</code>	Horizontal line, full width, at fractional height y .
<code>\mdLabel {x}{y}{text}</code>	Math-mode label at (x, y) .
<code>\mdLabel [color]{x}{y}{text}</code>	Same with filled node label
<code>\mdLabelRaw {x}{y}{text}</code>	Raw-text label at (x, y) .
<code>\mdLabelCenter {text}</code>	Math label at center (any aspect).
<code>\mdSpike {x}{y}{dir}</code>	L-corner; <code>dir</code> $\in \{tl, tr, bl, br\}$.
<code>\mdBlank</code>	Empty content (brackets only).
<code>\mdH</code>	Expands to the aspect value, i.e. the top edge's y -coordinate.

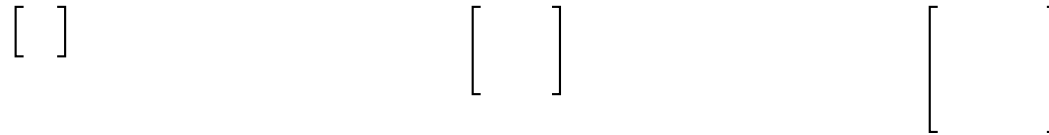
2 Main command

`\matrixdiagram[scale][aspect]{content}`

Inside `content`, the raw TikZ canvas runs $x \in [0, 1]$ and $y \in [0, \text{aspect}]$: `aspect` is literally the y -coordinate of the top edge, and `\mdH` expands to that value, so $(0, \text{\mdH})$ – $(1, \text{\mdH})$ is always the top edge, whatever the aspect. Every built-in positioning command (`\mdFill`, `\mdLabel`, `\mdLabelRaw`, `\mdHLine`, `\mdSpike`, and the ...From diagonal variants) instead takes y as a plain fraction of the height, $y \in [0, 1]$, and multiplies by `\mdH` internally — so `y=0.5` always means mid-height, regardless of aspect. Raw `\draw` commands written directly inside `content` see the unscaled canvas, so they must reference `\mdH` themselves, e.g. $(1, \text{\mdH})$ for the top-right corner.

2.1 Scale (aspect = 1)

`\matrixdiagram[1]{\mdBlank}` `\matrixdiagram[2]{\mdBlank}` `\matrixdiagram[3]{\mdBlank}`

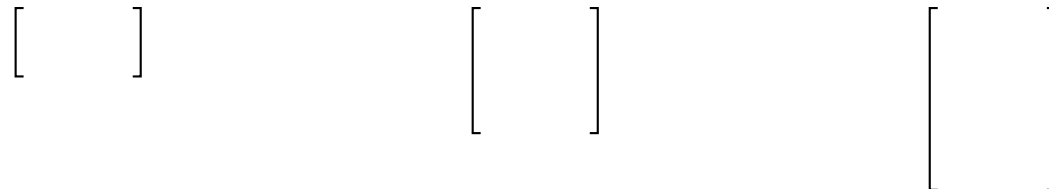


`\matrixdiagram[4]{\mdBlank}` `\matrixdiagram[5]{\mdBlank}`

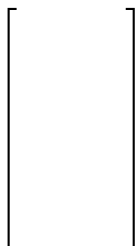


2.2 Aspect ratio (scale = 3)

`\matrixdiagram[3][0.5]{\mdBlank}` `\matrixdiagram[3][1]{\mdBlank}` `\matrixdiagram[3][1.5]{\mdBlank}`



```
$\matrixdiagram[3][2]{
  \mdBlank
}$
```



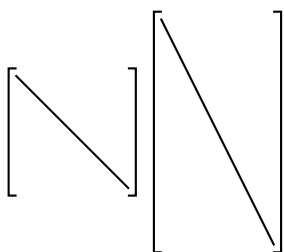
```
$\matrixdiagram[3][3]{
  \mdBlank
}$
```



3 Diagonal commands

3.1 `\mdDiag` — main diagonal

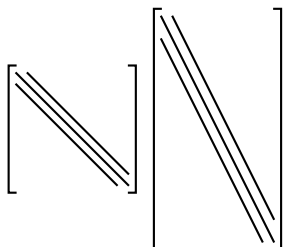
The single diagonal line from $(0, H)$ to $(1, 0)$.



```
$\matrixdiagram[3]{\mdDiag} \matrixdiagram[3][2]{\mdDiag}$
```

3.2 `\mdDiagBand` — tridiagonal band

Three parallel anti-diagonals with spacing $d = 0.1$.

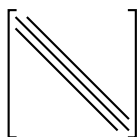


```
$\matrixdiagram[3]{\mdDiagBand} \matrixdiagram[3][2]{\mdDiagBand}$
```

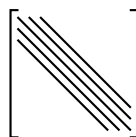
3.3 `\mdDiagBandN{n}` — variable band width

$n = 3, 5, 7, 9$ (spacing fixed at $d = 0.1$ per diagonal); `\mdDiagBandN{3}` is identical to `\mdDiagBand`:

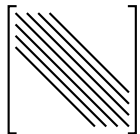
```
$\matrixdiagram[3]{\mdDiagBandN{3}}$
```



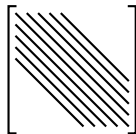
```
$\matrixdiagram[3]{\mdDiagBandN{5}}$
```



`\matrixdiagram[3]{\mdDiagBandN{7}}\$`

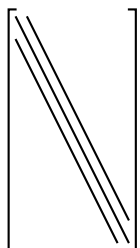


`\matrixdiagram[3]{\mdDiagBandN{9}}\$`

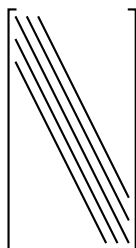


With `aspect = 2`:

`\matrixdiagram[3][2]{\mdDiagBandN{3}}\$`



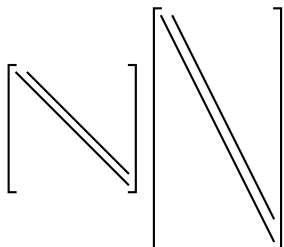
`\matrixdiagram[3][2]{\mdDiagBandN{5}}\$`



4 Bidiagonal commands

4.1 `\mdBidiagUpper` — main + super-diagonal

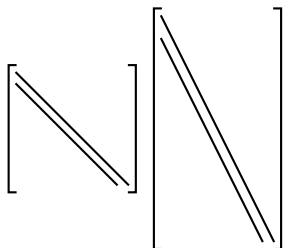
Main diagonal plus a single super-diagonal, offset $d = 0.1$ toward the top-right.



`\matrixdiagram[3]{\mdBidiagUpper} \matrixdiagram[3][2]{\mdBidiagUpper}\$`

4.2 `\mdBidiagLower` — main + sub-diagonal

Main diagonal plus a single sub-diagonal, offset $d = 0.1$ toward the bottom-left.

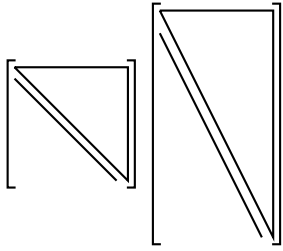


`\matrixdiagram[3]{\mdBidiagLower} \matrixdiagram[3][2]{\mdBidiagLower}\$`

5 Hessenberg command

5.1 `\mdHessenberg` — outline only (default)

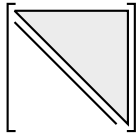
Upper-triangular outline $(0, H) \rightarrow (1, H) \rightarrow (1, 0)$ plus one sub-diagonal.



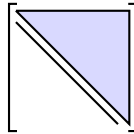
`\matrixdiagram[3]{\mdHessenberg}` `\matrixdiagram[3][2]{\mdHessenberg}`

5.2 `\mdHessenberg[color]` — with fill

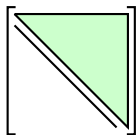
`\matrixdiagram[3]{\mdHessenberg[gray!15]}`



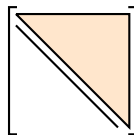
`\matrixdiagram[3]{\mdHessenberg[blue!15]}`



`\matrixdiagram[3]{\mdHessenberg[green!20]}`



`\matrixdiagram[3]{\mdHessenberg[orange!20]}`

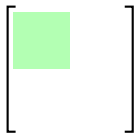


6 Fill and line commands

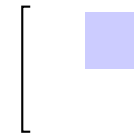
6.1 `\mdFill[color]{x0}{y0}{x1}{y1}`

The default color is gray!20:

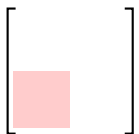
`\matrixdiagram[3]{`
`\mdFill[green!30]{0}{0.5}{0.5}{1}`
`}`



`\matrixdiagram[3]{`
`\mdFill[blue!20]{0.5}{0.5}{1}{1}`
`}`



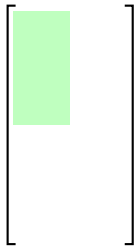
`\matrixdiagram[3]{`
`\mdFill[red!20]{0}{0}{0.5}{0.5}`
`}`



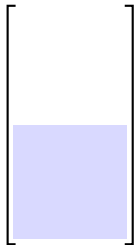
`\matrixdiagram[3]{`
`\mdFill{0}{0}{1}{1}`
`}`



With `aspect = 2` (y is still a fraction of the height, in $[0, 1]$):



`\matrixdiagram[3][2]{\mdFill[green!25]{0}{0.5}{0.5}{1}}$`

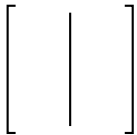


`\matrixdiagram[3][2]{\mdFill[blue!15]{0}{0}{1}{0.5}}$`

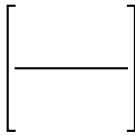
6.2 `\mdVLine{x}` and `\mdHLine{y}`

`\mdVLine{x}` runs the full height at $x \in [0, 1]$; `\mdHLine{y}` runs the full width at height $y \in [0, 1]$.

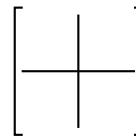
`\matrixdiagram[3]{
 \mdVLine{0.5}
}$`



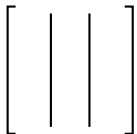
`\matrixdiagram[3]{
 \mdHLine{0.5}
}$`



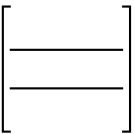
`\matrixdiagram[3]{
 \mdVLine{0.5}
 \mdHLine{0.5}
}$`



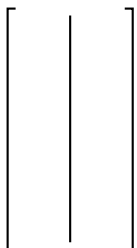
`\matrixdiagram[3]{
 \mdVLine{0.33}
 \mdVLine{0.67}
}$`



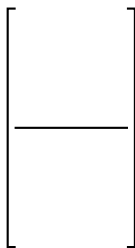
`\matrixdiagram[3]{
 \mdHLine{0.33}
 \mdHLine{0.67}
}$`



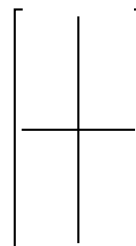
`\matrixdiagram[3][2]{
 \mdVLine{0.5}
}$`



`\matrixdiagram[3][2]{
 \mdHLine{0.5}
}$`



`\matrixdiagram[3][2]{
 \mdVLine{0.5}
 \mdHLine{0.5}
}$`



Combined with fills (block partitioning):

$$\left[\begin{array}{c|c} A & \\ \hline & D \end{array} \right]$$

```
\matrixdiagram[3]{
  \mdFill[blue!15]{0}{0.5}{0.5}{1}
  \mdFill[green!15]{0.5}{0}{1}{0.5}
  \mdVLine{0.5}
  \mdHLine{0.5}
  \mdLabel{0.25}{0.75}{A}
  \mdLabel{0.75}{0.25}{D}
}
```

$$\left[\begin{array}{c|c} & b \\ \hline A & \end{array} \right]$$

```
\matrixdiagram[3][2]{
  \mdFill[gray!15]{0}{0}{0.75}{1}
  \mdVLine{0.75}
  \mdLabel{0.375}{0.5}{A}
  \mdLabel{0.875}{0.5}{b}
}
```

7 Label commands

7.1 `\mdLabel{x}{y}{text}` — **math mode**

Places **text** in math mode at (x,y) ; the optional first argument fills the label's background node with that color.

```
\matrixdiagram[3]{
  \mdLabel{0.2}{0.5}{A}
}
```

$$\left[\begin{array}{c} A \end{array} \right]$$

```
\matrixdiagram[3]{
  \mdLabel[red!20]{0.5}{0.5}{Q^T}
}
```

$$\left[\begin{array}{c} Q^T \end{array} \right]$$

```
\matrixdiagram[3]{
  \mdLabel{0.5}{0.5}{\Sigma}
}
```

$$\left[\begin{array}{c} \Sigma \end{array} \right]$$

```
\matrixdiagram[3]{
  \mdDiagBand
  \mdLabel{0.25}{0.2}{T}
}
```

$$\left[\begin{array}{c} T \end{array} \right]$$

7.2 `\mdLabelCenter{text}` — **always centered**

Automatically placed at $(0.5, H/2)$ regardless of aspect ratio:

```

 $\matrixdiagram[3][1]{$ 
 $\mdLabelCenter{Q}$ 
 $\}$ 

```

$$\begin{bmatrix} Q \end{bmatrix}$$

```

 $\matrixdiagram[3][2]{$ 
 $\mdLabelCenter{Q}$ 
 $\}$ 

```

$$\begin{bmatrix} Q \end{bmatrix}$$

```

 $\matrixdiagram[3][3]{$ 
 $\mdLabelCenter{Q}$ 
 $\}$ 

```

$$\begin{bmatrix} Q \end{bmatrix}$$

```

 $\matrixdiagram[3][0.5]{$ 
 $\mdLabelCenter{Q}$ 
 $\}$ 

```

$$\begin{bmatrix} Q \end{bmatrix}$$

7.3 $\mdLabelRow{x}{y}{text}$ — raw text

Same placement as $\mdLabel$, but **text** is typeset as-is instead of in math mode — useful for plain words like “dense” or for text that already contains its own $\dots$.

$$\begin{bmatrix} \text{dense} \end{bmatrix} \begin{bmatrix} \approx 0 \end{bmatrix}$$

```

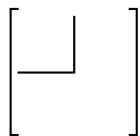
 $\matrixdiagram[3]{$ 
 $\mdLabelRow{0.5}{0.5}{dense}$ 
 $\}$ 
 $\matrixdiagram[3]{$ 
 $\mdLabelRow{0.5}{0.5}{\approx 0}$ 
 $\}$ 

```

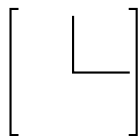
8 Spike commands

$\mdSpike{x}{y}{dir}$ draws an L-shaped corner at (x, y) . **tl** (top+left), **tr** (top+right), **bl** (bottom+left), **br** (bottom+right). Optionally, add a color fill via $\mdSpike[color]{x}{y}{dir}$.

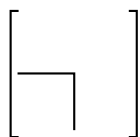
```
$\matrixdiagram[3]{
  \mdSpike{0.5}{0.5}{tl}
}$
```



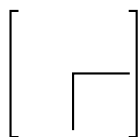
```
$\matrixdiagram[3]{
  \mdSpike{0.5}{0.5}{tr}
}$
```



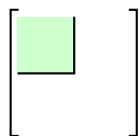
```
$\matrixdiagram[3]{
  \mdSpike{0.5}{0.5}{bl}
}$
```



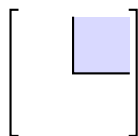
```
$\matrixdiagram[3]{
  \mdSpike{0.5}{0.5}{br}
}$
```



```
$\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
}$
```

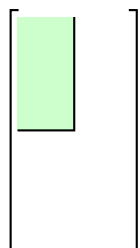


```
$\matrixdiagram[3]{
  \mdFill[blue!15]{0.5}{0.5}{1}{1}
  \mdSpike{0.5}{0.5}{tr}
}$
```

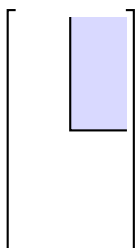


With aspect = 2, corner at mid-height (0.5, 0.5):

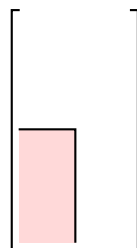
```
$\matrixdiagram[3][2]{
  \mdFill[green
    !20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
}$
```



```
$\matrixdiagram[3][2]{
  \mdFill[blue
    !15]{0.5}{0.5}{1}{1}
  \mdSpike{0.5}{0.5}{tr}
}$
```



```
$\matrixdiagram[3][2]{
  \mdFill[red
    !15]{0}{0}{0.5}{0.5}
  \mdSpike{0.5}{0.5}{bl}
}$
```

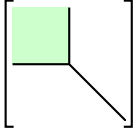


9 Clipped diagonal commands

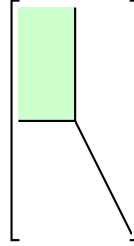
The ...From variants force the diagonal band from $[x_1, y_1]$ to $[x_2, y_2]$. Always pair with `\mdSpike` at the same corner.

9.1 `\mdDiagFrom{x1}{y1}{x2}{y2}`

```
$$\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagFrom{0.5}{0.5}{1}{0}
}$
```

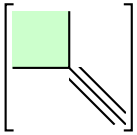


```
$$\matrixdiagram[3][2]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagFrom{0.5}{0.5}{1}{0}
}$
```

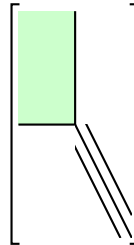


9.2 `\mdDiagBandFrom{x1}{y1}{x2}{y2}`

```
$$\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandFrom{0.5}{0.5}{1}{0}
}$
```



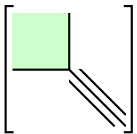
```
$$\matrixdiagram[3][2]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandFrom{0.5}{0.5}{1}{0}
}$
```



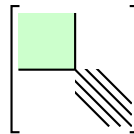
9.3 `\mdDiagBandNFrom{n}{x1}{y1}{x2}{y2}`

$n = 3, 5, 7$:

```
$$\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandNFrom{3}{0.5}{0.5}{1}{0}
}$
```



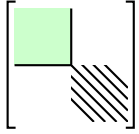
```
$$\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandNFrom{5}{0.5}{0.5}{1}{0}
}$
```



```

 $\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{t1}
  \mdDiagBandNFrom{7}{0.5}{0.5}{1}{0}
}$ 

```



10 Realistic equations

10.1 QR decomposition: $A = QR$

$$\begin{bmatrix} & & \\ A & & \\ & & \end{bmatrix} = \begin{bmatrix} & & \\ Q & & \\ & & \end{bmatrix} \begin{bmatrix} & & \\ R & & \\ & & \end{bmatrix}$$

```

 $\matrixdiagram[3][2]{\mdLabelCenter{A}}$ 
=
 $\matrixdiagram[3][2]{\mdLabelCenter{Q}}$ 
 $\matrixdiagram[3]{\mdLabelCenter{R}}$ 

```

10.2 Bidiagonalization: $A = UBV^T$

$$\begin{bmatrix} & & \\ A & & \\ & & \end{bmatrix} = \begin{bmatrix} & & \\ U & & \\ & & \end{bmatrix} \begin{bmatrix} & & \\ \diagdown & & \\ & \diagup & \end{bmatrix} \begin{bmatrix} & & \\ V^T & & \\ & & \end{bmatrix}$$

```

 $\matrixdiagram[3]{\mdLabelCenter{A}}$ 
=
 $\matrixdiagram[3]{\mdLabelCenter{U}}$ 
 $\matrixdiagram[3]{\mdBidiagUpper}$ 
 $\matrixdiagram[3]{\mdLabelCenter{V^T}}$ 

```

10.3 Similarity: $Q^T T_m Q = \tilde{T}_m$

$$\begin{bmatrix} & & \\ Q^T & & \\ & & \end{bmatrix} \begin{bmatrix} & & \\ \diagdown & & \\ & \diagup & \end{bmatrix} \begin{bmatrix} & & \\ Q & & \\ & & \end{bmatrix} = \begin{bmatrix} & & \\ \tilde{T}_m & & \\ & & \end{bmatrix}$$

```

 $\matrixdiagram[3]{\mdLabelCenter{Q^T}}$ 
 $\matrixdiagram[3]{
  \mdDiagBand
  \mdLabel{0.25}{0.15}{T_m}
}$ 
 $\matrixdiagram[3]{\mdLabelCenter{Q}}$ 
=
 $\matrixdiagram[3]{
  \mdFill[green!25]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{t1}
  \mdDiagBandFrom{0.5}{0.5}{1}{0}
  \mdLabel{0.3}{0.12}{\widetilde{T}_m}
}$ 

```

10.4 Hessenberg reduction: $Q^T A Q = H$

$$\left[\begin{array}{c} Q^T \end{array} \right] \left[\begin{array}{c} A \end{array} \right] \left[\begin{array}{c} Q \end{array} \right] = \left[\begin{array}{c} \text{H} \end{array} \right]$$

```

 $\matrixdiagram[3]{\mdLabelCenter{Q^T}}$ 
 $\matrixdiagram[3]{\mdLabelCenter{A}}$ 
 $\matrixdiagram[3]{\mdLabelCenter{Q}}$ 
=
 $\matrixdiagram[3]{$ 
 $\mdHessenberg$ 
 $\mdLabel{0.7}{0.55}{H}$ 
 $}$ 

```

10.5 Schur complement structure

$$\left[\begin{array}{cc} A & B \\ C & S \end{array} \right]$$

```

 $\matrixdiagram[4]{$ 
 $\mdFill[blue!10]{0}{0.5}{0.5}{1}$ 
 $\mdFill[green!10]{0.5}{0}{1}{0.5}$ 
 $\mdFill[gray!10]{0.5}{0.5}{1}{1}$ 
 $\mdVLine{0.5}$ 
 $\mdHLine{0.5}$ 
 $\mdLabel{0.25}{0.75}{A}$ 
 $\mdLabel{0.75}{0.75}{B}$ 
 $\mdLabel{0.25}{0.25}{C}$ 
 $\mdLabel{0.75}{0.25}{S}$ 
 $}$ 

```

11 Implementation

```

(*package)
\RequirePackage{tikz}
\RequirePackage{xparse}
\RequirePackage{calc}

\usetikzlibrary{arrows.meta, bending, matrix, positioning, calc, arrows}

%% -----
%% Internal dimensions
%% -----
\newdimen\md@bracewidth % length of the horizontal tick at each bracket corner
\newdimen\md@innersep % gap between the content and the brackets
\newdimen\md@baseunit % size of one coordinate unit at scale=1
\newdimen\md@hsep % margin added left and right of every \matrixdiagram
\md@bracewidth=3pt
\md@innersep=2pt
\md@baseunit=0.5cm
\md@hsep=3pt

%% \md@H: current aspect ratio = y-coordinate of top edge. Default 1.
\def\md@H{1}
%% \mdH: user-facing alias usable inside content arguments
\def\mdH{\md@H}

%% \md@bandstep: spacing between adjacent diagonals in a band, in x-units.
\def\md@bandstep{0.1}

%% -----
%% Core drawing macro: \md@draw@box{scale}{aspect}{content}
%% -----
\newcommand{\md@draw@box}[3]{%

```

```

\def\md@H{#2}%
\kern\md@hsep
\vcenter{\hbox{%
  \tikz[baseline={(\MDNODO.base)}]{%
    \node[inner sep=\md@innersep](MDNODO){%
      \tikz[baseline={(\MDcenter)-(0.25em,0.25em)$}],
        x={#1*\md@baseunit}, y={#1*\md@baseunit}]{%
          \coordinate (MDcenter) at (0.5,{#2*0.5});
          \draw[draw=none] (0,0) rectangle (1,{#2});
          #3
        }%
      };
    \draw[thick]
      (MDNODO.north west)+(\md@bracewidth,0)
      -- (MDNODO.north west)
      -- (MDNODO.south west)
      -- (\MDNODO.south west)+(\md@bracewidth,0$);
    \draw[thick]
      (MDNODO.north east)+(-\md@bracewidth,0)
      -- (MDNODO.north east)
      -- (MDNODO.south east)
      -- (\MDNODO.south east)+(-\md@bracewidth,0$);
    }%
  }}%
\kern\md@hsep
}

%% -----
%% \matrixdiagram[scale][aspect]{content}
%% -----
\NewDocumentCommand{\matrixdiagram}{0{1} 0{1} m}{%
  \md@draw@box{#1}{#2}{#3}%
}

%% -----
%% Content commands | all use \md@H for the top edge
%% -----

%% \md@superdiag{k} | super-diagonal offset by k*\md@bandstep
\newcommand{\md@superdiag}[1]{%
  \draw[thick] ({#1*\md@bandstep},\md@H) -- (1,{#1*\md@bandstep*\md@H});%
}

%% \md@subdiag{k} | sub-diagonal offset by k*\md@bandstep
\newcommand{\md@subdiag}[1]{%
  \draw[thick] (0,{\md@H-#1*\md@bandstep*\md@H}) -- ({1-#1*\md@bandstep},0);%
}

%% \mdDiagBandN{n} | n-diagonal band (n odd), spacing \md@bandstep between diagonals.
%% n=1 draws only the main diagonal. Max useful n is 9.
\newcommand{\mdDiagBandN}[1]{%
  \draw[thick] (0,\md@H) -- (1,0);%
  \pgfmathtruncatemacro{\md@half}{int((#1-1)/2)}%
  \ifnum\md@half>0
    \foreach \md@k in {1,...,\md@half}{%
      \md@subdiag{\md@k}%
      \md@superdiag{\md@k}%
    }
  \fi
}

```

```

    }%
  \fi
}

%% \mdDiag / main anti-diagonal (top-left to bottom-right)
\newcommand{\mdDiag}{\mdDiagBandN{1}}
\let\mdIdentity\mdDiag

%% \mdDiagBand / tridiagonal band
\newcommand{\mdDiagBand}{\mdDiagBandN{3}}

%% \mdFill[color]{x0}{y0}{x1}{y1} | y0,y1 are fractions of the height, in [0,1]
\newcommand{\mdFill}[5][gray!20]{%
  \fill[#1] (#2,{#3*\md@H}) rectangle (#4,{#5*\md@H});%
}

%% \mdLabel[fill]{x}{y}{text} | math-mode label, optional node fill color; y in [0,1]
\newcommand{\mdLabel}[4][none]{%
  \node[fill=#1] at (#2,{#3*\md@H}) {$#4$};%
}

%% \mdLabelRaw[fill]{x}{y}{text} | raw text label, optional node fill color; y in [0,1]
\newcommand{\mdLabelRaw}[4][none]{%
  \node[fill=#1] at (#2,{#3*\md@H}) {#4};%
}

%% \mdLabelCenter{text} | math label at the center (x=0.5, y=H/2)
\newcommand{\mdLabelCenter}[1]{%
  \node at (0.5,{0.5*\md@H}) {$#1$};%
}

%% \mdVLine{x} | full-height vertical line (0 to H)
\newcommand{\mdVLine}[1]{%
  \draw[thick] (#1,0) -- (#1,\md@H);%
}

%% \mdHLine{y} | full-width horizontal line; y is a fraction of the height, in [0,1]
\newcommand{\mdHLine}[1]{%
  \draw[thick] (0,{#1*\md@H}) -- (1,{#1*\md@H});%
}

%% -----
%% Clipped diagonal commands
%% Draw diagonal bands clipped to the rectangle (x1,y1)--(x2,y2),
%% where (x1,y1) is the bottom-left corner and (x2,y2) is the top-right.
%%
%% \mdDiagFrom{x1}{y1}{x2}{y2}
%% \mdDiagBandFrom{x1}{y1}{x2}{y2}
%% \mdDiagBandNFrom{n}{x1}{y1}{x2}{y2}
%% n=1 draws only the main diagonal (no off-diagonals).
%%
%% Typical spike usage | clip to the region outside the block:
%% \mdSpike{0.5}{0.5}{tl}
%% \mdDiagBandFrom{0.5}{0}{1}{0.5} % bottom-left=(0.5,0), top-right=(1,0.5)
%% -----

```

```

%% \mdDiagBandNFrom{n}{x1}{y1}{x2}{y2} | y1,y2 are fractions of the height, in [0,1]
\newcommand{\mdDiagBandNFrom}[5]{%
  \begin{scope}%
    \clip (#2,{#3*\md@H}) rectangle (#4,{#5*\md@H});%
    \mdDiagBandN{#1}%
  \end{scope}%
}

%% \mdDiagFrom{x1}{y1}{x2}{y2}
\newcommand{\mdDiagFrom}[4]{\mdDiagBandNFrom{1}{#1}{#2}{#3}{#4}}

%% \mdDiagBandFrom{x1}{y1}{x2}{y2}
\newcommand{\mdDiagBandFrom}[4]{\mdDiagBandNFrom{3}{#1}{#2}{#3}{#4}}

%% -----
%% Spike commands | L-shaped block boundary corners
%% tl: corner at (x,y), arms to top (y=H) and left (x=0)
%% tr: corner at (x,y), arms to top (y=H) and right (x=1)
%% bl: corner at (x,y), arms to bottom (y=0) and left (x=0)
%% br: corner at (x,y), arms to bottom (y=0) and right (x=1)
%% -----
\def\md@tl{tl}\def\md@tr{tr}\def\md@bl{bl}\def\md@br{br}

%% \mdSpike{x}{y}{dir} | y is a fraction of the height, in [0,1]
\newcommand{\mdSpike}[3]{%
  \def\md@spikedir{#3}%
  \ifx\md@spikedir\md@tl \draw[thick] (#1,\md@H) -- (#1,{#2*\md@H}) -- (0,{#2*\md@H});%
  \else\ifx\md@spikedir\md@tr \draw[thick] (#1,\md@H) -- (#1,{#2*\md@H}) -- (1,{#2*\md@H});%
  \else\ifx\md@spikedir\md@bl \draw[thick] (#1,0) -- (#1,{#2*\md@H}) -- (0,{#2*\md@H});%
  \else\ifx\md@spikedir\md@br \draw[thick] (#1,0) -- (#1,{#2*\md@H}) -- (1,{#2*\md@H});%
  \fi\fi\fi\fi
}

%% -----
%% Bidiagonal commands
%% upper: main diagonal + one super-diagonal
%% lower: main diagonal + one sub-diagonal
%% -----

%% \mdBidiagUpper | main diagonal + super-diagonal
\newcommand{\mdBidiagUpper}{%
  \draw[thick] (0,\md@H) -- (1,0);%
  \md@superdiag{1}%
}

%% \mdBidiagLower | main diagonal + sub-diagonal
\newcommand{\mdBidiagLower}{%
  \draw[thick] (0,\md@H) -- (1,0);%
  \md@subdiag{1}%
}

%% -----
%% Hessenberg command
%% \mdHessenberg[fillcolor]
%% No argument: outline only (default).
%% With color argument: filled triangle + outline.

```

```

%% -----
\newcommand{\mdHessenberg}[1][none]{%
  \def\md@hessarg{#1}\def\md@hessnone{none}%
  \ifx\md@hessarg\md@hessnone
    \draw[thick] (0,\md@H) -- (1,\md@H) -- (1,0) -- (0,\md@H);%
  \else
    \fill[#1] (0,\md@H) -- (1,\md@H) -- (1,0) -- cycle;%
    \draw[thick] (0,\md@H) -- (1,\md@H) -- (1,0) -- (0,\md@H);%
  \fi
  \md@subdiag{1}%
}

%% \mdBlank / empty content (brackets only)
\newcommand{\mdBlank}{}
\end{package}

```